Systematic Search for Singularities and Instabilities in Euler Flows

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GFD Program @ Woods Hole Oceanographic Institution

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Agenda

PART I:

Systematic Search for Singularities via PDE Optimization

Joint work with:

- Di Kang (former post-doc, now in industry)
- Xinyu Zhao (post-doc, Assistant Professor at the New Jersey Institute of Technology as of September 2024)

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Maximum amplification of enstrophy in three-dimensional Navier–Stokes flows

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Searching for Singularities in Navier–Stokes Flows Based on the Ladyzhenskaya–Prodi–Serrin Conditions

Di Kang[1] · Bartosz Protas[1]

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Systematic Search for Singularities in 3D Euler Flows

Xinyu Zhao[1] · Bartosz Protas[1]

- Navier-Stokes system ($\Omega = [0, L]^d$, $d = 2, 3$)

$$\begin{cases} \dfrac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \boldsymbol{\nabla})\mathbf{u} + \boldsymbol{\nabla} p - \nu \Delta \mathbf{u} = \mathbf{0}, & \text{in } \Omega \times (0, T] \\ \boldsymbol{\nabla} \cdot \mathbf{u} = 0, & \text{in } \Omega \times (0, T] \\ \mathbf{u} = \mathbf{u}_0 & \text{in } \Omega \text{ at } t = 0 \\ \text{Periodic Boundary Conditions} & \text{on } \Gamma \times (0, T] \end{cases}$$

- **The Big Question:**

  Given a smooth initial condition $\mathbf{u}_0$, does the Navier-Stokes system always admit smooth solutions $\mathbf{u}(t)$ for arbitrarily long times t?

- One of the Clay Institute "Millennium Problems" ($ 1M prize!)

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- What could go wrong with solutions to the Navier-Stokes equation?
- Consider its vorticity formulation ($\omega = \nabla \times \mathbf{u}$)

$$\frac{\partial \omega}{\partial t} + (\mathbf{u} \cdot \nabla)\omega = \frac{d\omega}{dt} = \nu \Delta \omega + \underbrace{(\nabla \mathbf{u})\,\omega}_{\text{“vortex stretching”}}$$

- Velocity $\mathbf{u}$ is obtained from vorticity using the Biot-Savart kernel $\mathbf{G}$

$$\mathbf{u} = \nabla \Delta^{-1} \omega = \int_\Omega \mathbf{G}(\cdot, \mathbf{x}')\omega(\mathbf{x}')\, d\mathbf{x}' = \mathbf{G} * \omega$$

- The vorticity equation has a quadratic source term (assume $\nu = 0$)

$$\frac{d\omega}{dt} = [\nabla(\mathbf{G} * \omega)]\, \omega$$

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- What could this imply?

- Consider a (very) simple ODE model problem

$$\frac{dy}{dt} = y^2, \; y(0) = y_0 \quad \text{with solution} \quad y(t) = \frac{y_0}{1 - y_0\, t}$$

- The solution $y(t)$ becomes unbounded ("blows up") as $t \to t_0 = \frac{1}{y_0}$

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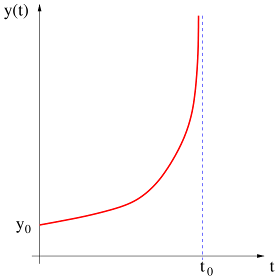

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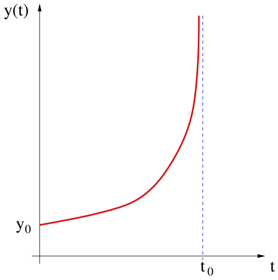

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- Another simple model problem — inviscid Burgers equation

$$\begin{aligned} \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} &= 0 \qquad \text{for } t > 0,\ x \in \mathbb{R}, \\ u(0,x) &= \phi(x) \quad \text{for } x \in \mathbb{R} \end{aligned}$$

  with (implicit) solution $u(t,x) = \phi(x - u(t,x)\,t)$

- The solution $u(t,x)$ develops a shock and becomes non-differentiable at time $t \to t_0 = \frac{-1}{\min_x \frac{d\phi(x)}{dx}}$

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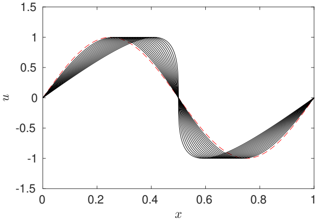

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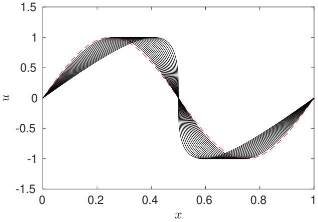

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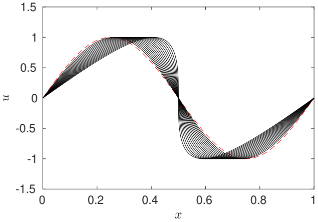

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- Can such singular behavior arise in the Navier-Stokes system in finite time?
- Who cares?
  - Well, if its solutions can become singular, then the Navier-Stokes system is not a correct model for viscous incompressible fluids and must be amended (by modifying the viscous terms)
- 2D Case
  - Existence theory complete — smooth and unique solutions exist for arbitrary times and arbitrarily large data
- 3D Case
  - Weak solutions (possibly nonsmooth) exist for arbitrary times
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The Enstrophy Condition

- A Key Quantity — Enstrophy

$$\mathcal{E}(t) \triangleq \int_\Omega |\boldsymbol{\nabla} \times \mathbf{u}|^2 \, d\Omega \qquad (= \|\boldsymbol{\nabla}\mathbf{u}\|_2^2)$$

- Smoothness of Solutions $\Longleftrightarrow$ Bounded Enstrophy (Foias & Temam, 1989)

$$\max_{t\in[0,T]} \mathcal{E}(t) < \infty \quad ???$$

- Can estimate $\frac{d\mathcal{E}(t)}{dt}$ using the momentum equation, Sobolev's embeddings, Young and Cauchy-Schwartz inequalities, ...
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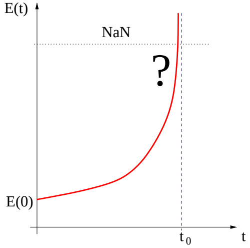

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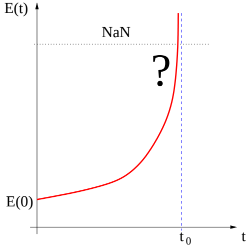

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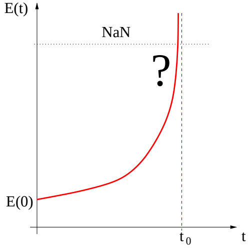

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- Bounds on the rate of growth of enstrophy — general form

$$\frac{d\mathcal{E}}{dt} < C\,\mathcal{E}^{\alpha}, \quad C > 0, \quad \alpha = \alpha(d) > 0$$

- Energy equation ($\mathcal{K}(t) \triangleq \int_\Omega \mathbf{u}^2 \, d\Omega$)

$$\frac{d\mathcal{K}}{dt} = -2\nu\mathcal{E}$$

- When $\alpha <= 2$, by Grönwall's inequality: $\mathcal{E}(t) \leq \mathcal{E}_0 \exp\left[\frac{C\mathcal{K}_0}{2\nu}\right]$
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- Bounds on the rate of growth of enstrophy — general form

$$\frac{d\mathcal{E}}{dt} < C\,\mathcal{E}^{\alpha}, \quad C > 0, \quad \alpha = \alpha(d) > 0$$

- Energy equation ($\mathcal{K}(t) \triangleq \int_\Omega \mathbf{u}^2 \, d\Omega$)

$$\frac{d\mathcal{K}}{dt} = -2\nu\mathcal{E}$$

$$\mathcal{K}(t) - \mathcal{K}(0) = -2\nu \int_0^t \mathcal{E}(\tau)\, d\tau \quad \Longrightarrow \quad \int_0^t \mathcal{E}(\tau)\, d\tau \leq \frac{1}{2\nu}\mathcal{K}_0$$

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$$\frac{d\mathcal{E}(t)}{dt} \le \frac{27C^2}{128\nu^3}\mathcal{E}(t)^3$$

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On the Nature of Possible Blow-up

- As the hypothetical blow-up time t_0 is approached ...
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- Local well-posedness of classical solutions in Sobolev spaces

Theorem (Kato, 1972)

If $\mathbf{u}_0 \in H^m(\mathbb{T}^3)$ or $H^m(\mathbb{R}^3)$ for some $m > 5/2$ and satisfies $\boldsymbol{\nabla} \cdot \mathbf{u}_0 = 0$, then there exists a time $T > 0$ such that the Euler system has a unique solution $\mathbf{u} = \mathbf{u}(t, \mathbf{x})$ with the initial condition $\mathbf{u}_0$ and the solution satisfies $\mathbf{u} \in C([0, T]; H^m) \bigcap C^1([0, T]; H^{m-1})$.

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- QUESTION: How to look for initial data $\mathbf{u}_0$ which may lead to singularity in finite time?

- The computational search for singularities in Euler flows has had a long history: Brachet et al. (1983), Pumir & Siggia (1990), Brachet (1991), Kerr (1993), Pelz (2001), Bustamante & Kerr (2008), Matsumoto et al. (2008), Ohkitani & Constantin (2008), Ohkitani (2008), Grafke et al. (2008), Gibbon et al. (2008), Hou (2009), Siegel & Caflisch (2009),Orlandi et al. (2012), Bustamante & Brachet (2012), Kerr (2013), Orlandi et al. (2014), Campolina & Mailybaev (2018), Larios et al. (2018),...

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- Want to maximize enstrophy at time T, with $\mathcal{E}_0 := \mathcal{E}(\mathbf{u}_0) > 0$ fixed, to see if $\mathcal{E}_T(\mathbf{u}_0) := \mathcal{E}(\mathbf{u}(T; \mathbf{u}_0))$ can become infinite

Problem (1)

$$\max_{\mathbf{u}_0 \in \mathcal{Q}_{\mathcal{E}_0}} \mathcal{E}_T(\mathbf{u}_0), \quad \textit{where}$$

$$\mathcal{Q}_{\mathcal{E}_0} = \left\{ \mathbf{u}_0 \in H^1(\Omega) : \; \boldsymbol{\nabla} \cdot \mathbf{u}_0 = 0, \; \int_\Omega \mathbf{u}_0 \, d\mathbf{x} = 0, \; \mathcal{E}(\mathbf{u}_0) = \mathcal{E}_0 \right\},$$

$$\textit{subject to:} \quad \begin{cases} \dfrac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \boldsymbol{\nabla})\mathbf{u} + \boldsymbol{\nabla} p - \nu \Delta \mathbf{u} = \mathbf{0}, & \textit{in } \Omega \times (0, T] \\ \boldsymbol{\nabla} \cdot \mathbf{u} = 0, & \textit{in } \Omega \times (0, T] \\ \mathbf{u} = \mathbf{u}_0 & \textit{in } \Omega \textit{ at } t = 0 \\ \textit{Periodic Boundary Conditions} & \textit{on } \Gamma \times (0, T] \end{cases}$$

- A formidable, but solvable, PDE optimization problem

- Want to maximize enstrophy at time T, with $\mathcal{E}_0 := \mathcal{E}(\mathbf{u}_0) > 0$ fixed, to see if $\mathcal{E}_T(\mathbf{u}_0) := \mathcal{E}(\mathbf{u}(T;\mathbf{u}_0))$ can become infinite

Problem (1)

$$\max_{\mathbf{u}_0 \in \mathcal{Q}_{\mathcal{E}_0}} \mathcal{E}_T(\mathbf{u}_0), \quad \textit{where}$$

$$\mathcal{Q}_{\mathcal{E}_0} = \left\{ \mathbf{u}_0 \in H^1(\Omega) : \ \boldsymbol{\nabla} \cdot \mathbf{u}_0 = 0, \ \int_\Omega \mathbf{u}_0 \, d\mathbf{x} = 0, \ \mathcal{E}(\mathbf{u}_0) = \mathcal{E}_0 \right\},$$

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- A formidable, but solvable, PDE optimization problem

- *Local* maximizers found via discretized gradient flow

$$\mathbf{u}_{0;\mathcal{E}_0,T}^{(n+1)} = \mathbb{P}_{\mathcal{Q}_{\mathcal{E}_0}} \left(\mathbf{u}_{0;\mathcal{E}_0,T}^{(n)} + \tau_n \nabla \mathcal{E}_T \left(\mathbf{u}_{0;\mathcal{E}_0,T}^{(n)} \right) \right),$$
$$\mathbf{u}_{0;\mathcal{E}_0,T}^{(1)} = \mathbf{u}^0,$$

where:

- $\nabla \mathcal{E}_T(\mathbf{u}_0)$ is the gradient of the objective functional $\mathcal{E}_T(\mathbf{u}_0)$ with respect to the initial data $\mathbf{u}_0$
- step size $\tau^{(n)}$ is found via *arc minimization* and the projection on the constraint manifold $\mathcal{Q}_{\mathcal{E}_0}$ is given by

$$\mathbb{P}_{\mathcal{Q}_{\mathcal{E}_0}}(\mathbf{u}_0) = \sqrt{\frac{\mathcal{E}_0}{\mathcal{E}_T(\mathbf{u}_0)}}\, \mathbf{u}_0$$

- *Local* maximizers found via discretized gradient flow

$$
\begin{aligned}
\mathbf{u}^{(n+1)}_{0;\mathcal{E}_0,T} &= \mathbb{P}_{\mathcal{Q}_{\mathcal{E}_0}}\left(\mathbf{u}^{(n)}_{0;\mathcal{E}_0,T} + \tau_n \nabla\mathcal{E}_T\left(\mathbf{u}^{(n)}_{0;\mathcal{E}_0,T}\right)\right), \\
\mathbf{u}^{(1)}_{0;\mathcal{E}_0,T} &= \mathbf{u}^0,
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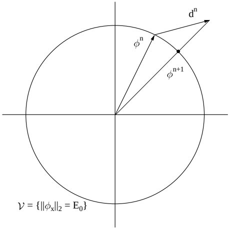

- How to ensure the required smoothness of the gradients $\nabla\mathcal{E}_T \in H^1$?
- Defining the from *adjoint system*

$$\mathcal{L}^* \begin{bmatrix} \mathbf{u}^* \\ p^* \end{bmatrix} := \begin{bmatrix} -\partial_t \mathbf{u}^* - \left[\boldsymbol{\nabla}\mathbf{u}^* + \boldsymbol{\nabla}\mathbf{u}^{*^T}\right]\mathbf{u} - \boldsymbol{\nabla} p^* - \nu\boldsymbol{\Delta}\mathbf{u}^* \\ -\boldsymbol{\nabla}\cdot\mathbf{u}^* \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Delta}\mathbf{u} \\ 0 \end{bmatrix},$$

$$\mathbf{u}^*(T) = \mathbf{0}$$

the Gâteaux differential of $\mathcal{E}_T(\mathbf{u}_0)$ becomes $\mathcal{E}_T'(\mathbf{u}_0;\mathbf{u}_0') = \int_\Omega \mathbf{u}_0'\cdot\mathbf{u}^*(0)\,d\mathbf{x}$

- Since $\mathcal{E}_T'(\mathbf{u}_0,\cdot)$ is a bounded linear functional on $L^2(\Omega)$ and on $H^1(\Omega)$, the gradient can be deduced from the Riesz representation theorem

$$\mathcal{E}_T'(\mathbf{u}_0;\mathbf{u}_0') = \left\langle \nabla^{L_2}\mathcal{E}_T(\mathbf{u}_0), \mathbf{u}_0' \right\rangle_{L^2(\Omega)} = \left\langle \nabla\mathcal{E}_T(\mathbf{u}_0), \mathbf{u}_0' \right\rangle_{H^1(\Omega)},$$

- Using the L^2 inner product: $\nabla^{L^2}\mathcal{E}_T(\mathbf{u}_0) = \mathbf{u}^*(0)$
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$$\left[\mathrm{Id} - \ell_1^2\,\boldsymbol{\Delta}\right]\nabla\mathcal{E}_T(\mathbf{u}_0) = \nabla^{L_2}\mathcal{E}_T(\mathbf{u}_0) \qquad \text{in } \Omega$$

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Computational Algorithm

- set $\mathcal{E}_0$ and T
- provide initial guess for the initial data $\mathbf{u}_{0;\mathcal{E}_0,T}$

1. solve the Navier-Stokes system for $\{\mathbf{u}, p\}$
2. solve the adjoint Navier-Stokes system for $\{\mathbf{u}^*, p^*\}$
3. use $\mathbf{u}$ and $\mathbf{u}^*$ to compute $\nabla^{L_2}\mathcal{E}_T$
4. determine the Sobolev gradient $\nabla^{H^1}\mathcal{E}_T$
5. update the initial data while enforcing the enstrophy constraint

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- iterate 1. through 5. until convergence, i.e. until

$$\frac{\mathcal{E}(\mathbf{u}^{(n+1)}_{0;\mathcal{E}_0,T}) - \mathcal{E}(\mathbf{u}^{(n)}_{0;\mathcal{E}_0,T})}{\mathcal{E}(\mathbf{u}^{(n)}_{0;\mathcal{E}_0,T})} < \epsilon$$

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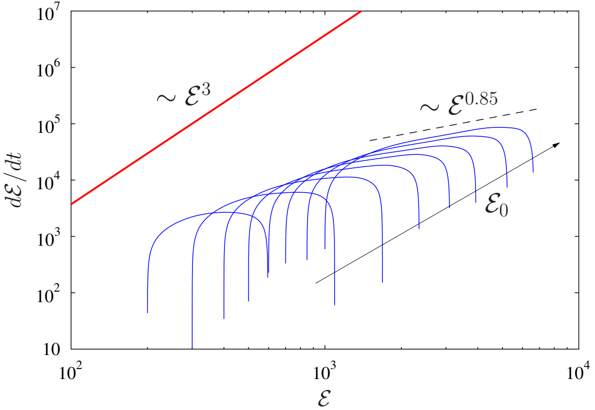
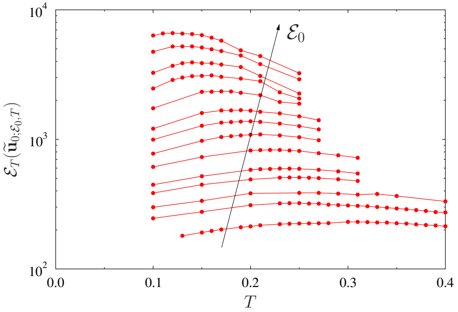
Maximum Sustained Rate of Enstrophy Growth $\frac{d\mathcal{E}}{dt} \sim C\,\mathcal{E}^{\alpha}$



extreme trajectories with optimal initial data $\widetilde{\mathbf{u}}_{0;\mathcal{E}_0,T}$

instantaneous maximizers $\widetilde{\mathbf{u}}_{\mathcal{E}_0}$

Maximum enstrophy $\max_{\mathbf{u}_0} \mathcal{E}(T)$ versus T for and $\mathcal{E}_0$



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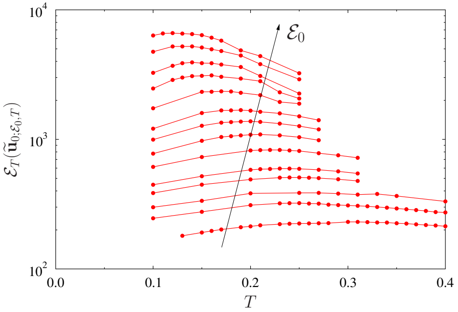
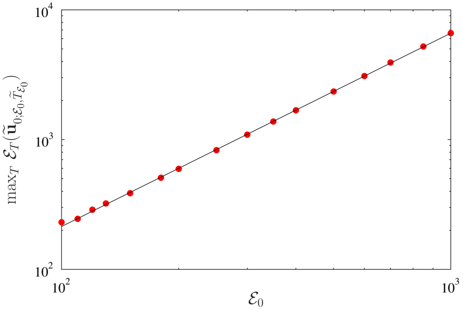
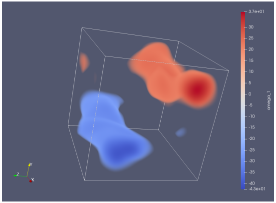


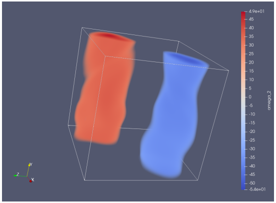


$$\boxed{\max_{T} \max_{t\in[0,T]} \mathcal{E}(\widetilde{\mathbf{u}}_{0;\mathcal{E}_0,T}(t)) \sim (0.224 \pm 0.006)\, \mathcal{E}_0^{1.490 \pm 0.004}}$$

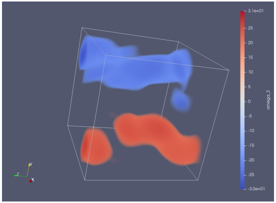
Structure of the optimal initial data $\widetilde{\mathbf{u}}_{0;\mathcal{E}_0,T}$ $(\mathcal{E}_0 = 500,\ T = 0.017)$



(a) ω_x



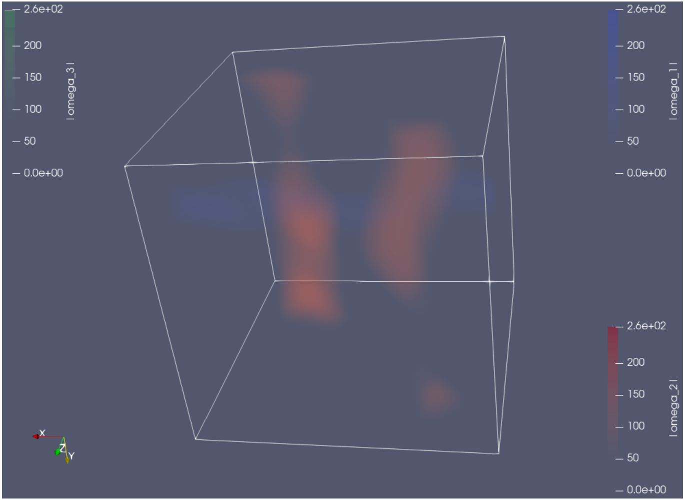
(b) ω_y



(c) ω_z

Time evolution of the extremal flow

($\mathcal{E}_0 = 500$ and $\widetilde{T}_{\mathcal{E}_0} = 0.17$)



- Maximize the Ladyzhenskaya-Prodi-Serrin functional with $q = 4$, $p = 8$

$$\Phi_T(\mathbf{u}_0) := \frac{1}{T}\int_0^T \|\mathbf{u}(\tau)\|_{L^4(\Omega)}^8 \, d\tau$$

Problem (2)

$$\max_{\mathbf{u}_0 \in \mathcal{L}_B} \Phi_T(\mathbf{u}_0), \quad \textit{where}$$

$$\mathcal{L}_B = \left\{ \mathbf{u}_0 \in H^{3/4}(\Omega) : \; \boldsymbol{\nabla} \cdot \mathbf{u}_0 = 0, \; \int_\Omega \mathbf{u}_0 \, d\mathbf{x} = 0, \; \|\mathbf{u}_0\|_{L^4(\Omega)} = B \right\},$$

$$\textit{subject to:} \quad \begin{cases} \dfrac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \boldsymbol{\nabla})\mathbf{u} + \boldsymbol{\nabla} p - \nu \Delta \mathbf{u} = \mathbf{0}, & \textit{in } \Omega \times (0, T] \\ \boldsymbol{\nabla} \cdot \mathbf{u} = 0, & \textit{in } \Omega \times (0, T] \\ \mathbf{u} = \mathbf{u}_0 & \textit{in } \Omega \textit{ at } t = 0 \\ \textit{Periodic Boundary Conditions} & \textit{on } \Gamma \times (0, T] \end{cases}$$

- Solutions sought in $H^{3/4}$, the largest Sobolev space with Hilbert structure embedded in L^4

- Maximize the Ladyzhenskaya-Prodi-Serrin functional with $q = 4$, $p = 8$

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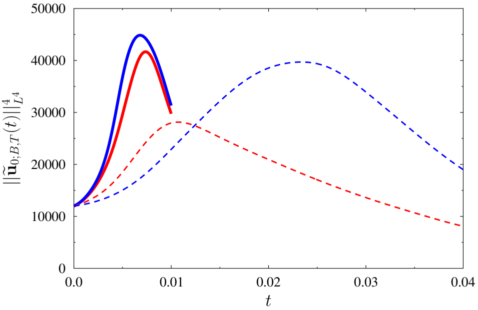
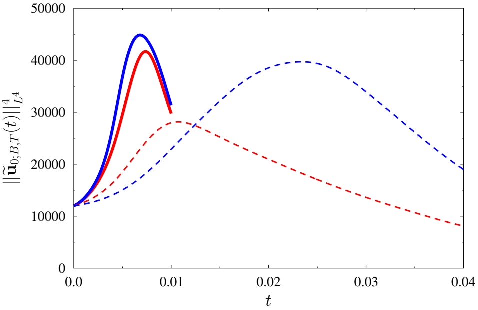
$\|\mathbf{u}(t)\|_{L^4}$ versus time t $\quad$ $\max_{\mathbf{u}_0} \Phi_T(\mathbf{u}_0)$ versus T

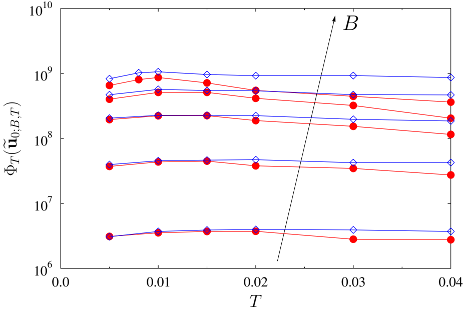


partially symmetric branch $\quad$ asymmetric branch

No evidence for unbounded growth of $\|\mathbf{u}(t)\|_{L^4}$ and singularity formation

$\|\mathbf{u}(t)\|_{L^4}$ versus time t $\quad$ $\max_{\mathbf{u}_0} \Phi_T(\mathbf{u}_0)$ versus T



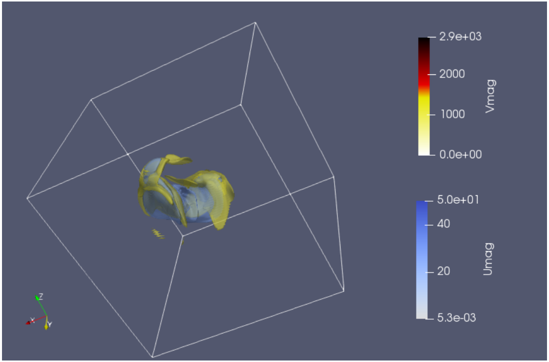


partially symmetric branch $\quad$ asymmetric branch

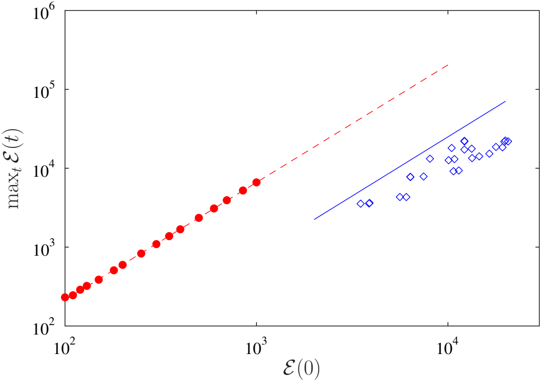
No evidence for unbounded growth of $\|\mathbf{u}(t)\|_{L^4}$ and singularity formation

Time evolution of the extremal flow

($B^4 = 12{,}000$ and $T = 0.01$)



Maximum enstrophy $\max_T \max_{\mathbf{u}_0} \mathcal{E}(T)$ vs. $\mathcal{E}_0$



solutions of Problem 1 — solutions of Problem 2

$$\boxed{\max_T \max_{\mathbf{u}_0} \mathcal{E}(T) \sim \mathcal{E}_0^{3/2}}$$

- The symmetry $\lambda \mathbf{u}_\lambda(\lambda t, \mathbf{x}) = \mathbf{u}(t, \mathbf{x}), \ \lambda \neq 0$ implies that the "size" of $\mathbf{u}_0$ and the time T and not independent
  - either $||\mathbf{u}_0||_{\dot{H}^3}$ or T can be fixed
- Set $m = 3$ in Kato's theorem and maximize

$$\Phi_T(\mathbf{u}_0) = ||\mathbf{u}(T; \mathbf{u}_0)||_{\dot{H}^3} \qquad \text{subject to} \qquad ||\mathbf{u}_0||_{\dot{H}^3} = 1$$

for different (increasing) T

- Unlike the Navier-Stokes system, the Euler system *does not* instantaneously regularize solutions (at $t = 0^+$)
  - to facilitate numerical solution of the Euler system, the optimal initial condition is sought in the Gevrey space G^σ of analytic functions endowed with the inner product

$$\langle \mathbf{v}, \mathbf{u} \rangle_{G^\sigma} = \sum_{\mathbf{j} \in \mathbb{Z}^3} (1 + |2\pi \mathbf{j}|^2)^3 e^{4\pi\sigma|\mathbf{j}|} \hat{\mathbf{v}}_{\mathbf{j}} \cdot \overline{\hat{\mathbf{u}}_{\mathbf{j}}}, \qquad \sigma > 0$$

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- Solution of the forward Euler and the backward adjoint system facilitated by time-reversibility

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$\widetilde{\boldsymbol{\eta}}_T^N$ — optimal initial condition obtained at time T with resolution N

- "short" times T: $\lim_{N\to\infty} \Phi_T\left(\widetilde{\boldsymbol{\eta}}_T^N\right) < \infty$

  $\Longrightarrow$ Euler system is well-posed on $[0, T]$

- "long" times T: $\lim_{N\to\infty} \Phi_T\left(\widetilde{\boldsymbol{\eta}}_T^N\right) = \infty$

  $\Longrightarrow$ a singularity possible at $t_0 \le T$

- In practice, we use $N = 128, 256, 512, 1024$

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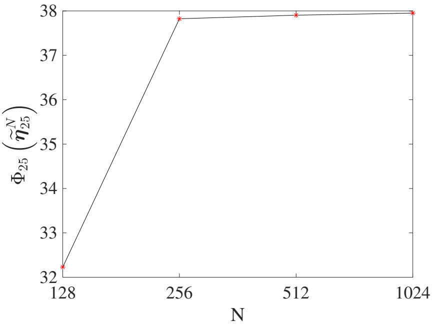
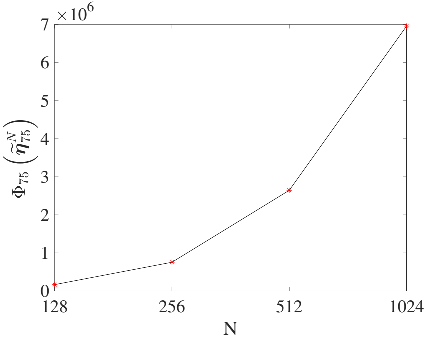
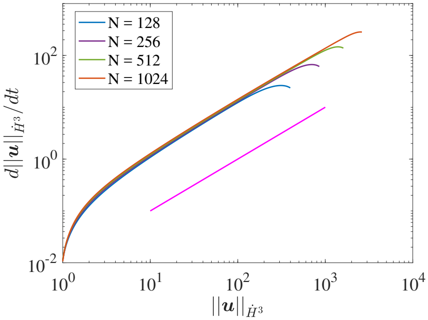


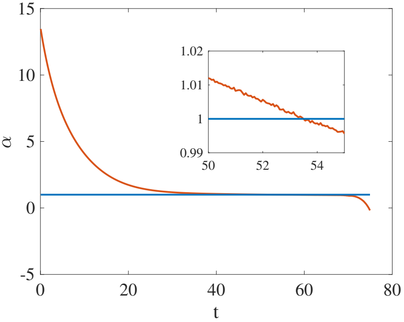
(a) $T = 25$



(b) $T = 75$

Objective functional $\Phi_T\left(\widetilde{\boldsymbol{\eta}}_T^N\right)$ obtained for different time windows T with different spatial resolutions N

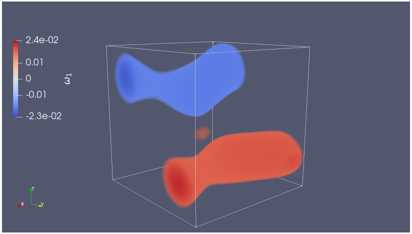




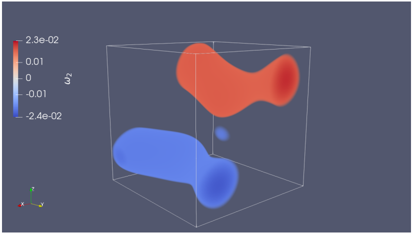
The growth rate of $\|\mathbf{u}(t)\|_{\dot H^3}$:

$$\frac{d\|\mathbf{u}(t)\|_{\dot H^3}}{dt} = C\|\mathbf{u}(t)\|_{\dot H^3}^{\alpha} \quad \Longrightarrow \quad \ln\left(\frac{d\|\mathbf{u}(t)\|_{\dot H^3}}{dt}\right) = \ln(C) + \alpha \ln\left(\|\mathbf{u}(t)\|_{\dot H^3}\right)$$

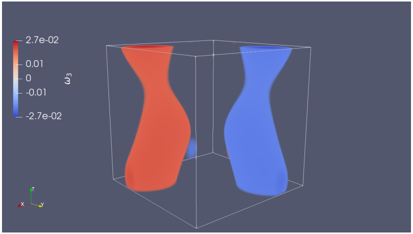
Computations become under-resolved at $t \approx 51.25$



(a) ω_x

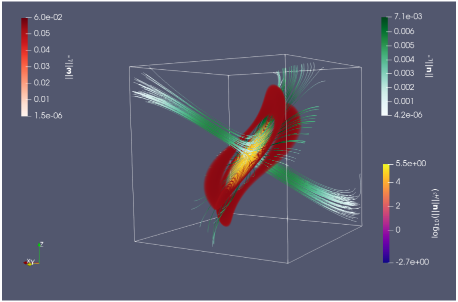


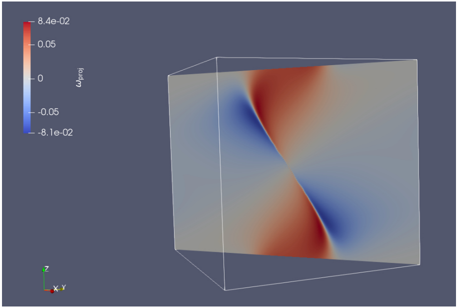
(b) ω_y



(c) ω_z

Vorticity components of the optimal initial condition $\widetilde{\boldsymbol{\eta}}_{75}^{1024}$





The vorticity field $\boldsymbol{\nabla} \times \mathbf{u}\left(75; \widetilde{\boldsymbol{\eta}}_{75}^{1024}\right)$ in the Euler flow corresponding to the optimal initial condition $\widetilde{\boldsymbol{\eta}}_{75}^{1024}$

Conclusions

- In the extreme Navier-Stokes flows the enstrophy $\mathcal{E}(t)$ and the norm $\|\mathbf{u}(t)\|_{L^4}$ remain finite at all times
  - hence, even in such worst-case scenario there is no evidence for formation of singularity in finite time
  - however, we do not know if the maximizers found are global
  - the extreme behavior in the two cases is realized by entirely different mechanisms
  - the scaling of the maximum growth of enstrophy with $\mathcal{E}_0$ is the same in both cases and, remarkably, the same as in 1D Burgers flows
- The extreme Euler flows
  - remain regular during sufficiently short times ($0 \leq t \leq 25$)
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    $\Longrightarrow$ the singular flow structure is nearly axisymmetric!

Summary of Relevant Energy-type Estimates

	Best Estimate	Sharp?
1D Burgers instantaneous	$\frac{d\mathcal{E}(t)}{dt} \leq \frac{3}{2}\left(\frac{1}{\pi^2\nu}\right)^{1/3}\mathcal{E}(t)^{5/3}$	Yes Lu & Doering (2008)
1D Burgers finite–time	$\max_{t\in[0,T]}\mathcal{E}(t) \leq C\,\mathcal{E}_0^{3/2}$	Yes Albritton & De Nitti (2023)
2D Navier–Stokes instantaneous	$\frac{d\mathcal{P}(t)}{dt} \leq -\left(\frac{\nu}{\mathcal{E}}\right)\mathcal{P}^2 + C_1\left(\frac{\mathcal{E}}{\nu}\right)\mathcal{P}$ $\frac{d\mathcal{P}(t)}{dt} \leq \frac{C_2}{\nu}\mathcal{K}^{1/2}\,\mathcal{P}^{3/2}$	Yes Ayala & P. (2013) Ayala, Doering & Simon (2017)
2D Navier–Stokes finite–time	$\max_{t>0}\mathcal{P}(t) \leq \left[\mathcal{P}_0^{1/2} + \frac{C_2}{4\nu^2}\mathcal{K}_0^{1/2}\mathcal{E}_0\right]^2$	Yes Ayala & P. (2013)
3D Navier–Stokes instantaneous	$\frac{d\mathcal{E}(t)}{dt} \leq \frac{27C^2}{128\nu^3}\mathcal{E}(t)^3$	Yes Lu & Doering (2008)
3D Navier–Stokes finite–time	$\mathcal{E}(t) \leq \frac{\mathcal{E}(0)}{\sqrt{1-4\frac{C\mathcal{E}(0)^2}{\nu^3}t}}$	No (???) Kang, Yun & P, (2020)

PART II:

Inviscid instabilities of the 2D Taylor-Green vortex

Joint work with:

- Xinyu Zhao (post-doc, Assistant Professor at the New Jersey Institute of Technology as of September 2024)
- Roman Shvydkoy (University of Illinois at Chicago)

J. Fluid Mech. (2024), *vol.* 999, A64, doi:10.1017/jfm.2024.946

JFM PAPERS

On the inviscid instability of the 2-D Taylor–Green vortex

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- 2D Euler system on a periodic domain

$$\partial_t \omega + (\boldsymbol{u} \cdot \nabla)\omega = 0, \qquad (\boldsymbol{x}, t) \in \mathbb{T}^2 \times (0, T],$$
$$\omega\big|_{t=0} = \omega_0, \qquad \boldsymbol{x} \in \mathbb{T}^2,$$

  where
  - $\omega = \nabla \times \boldsymbol{u}$ is the scalar vorticity field satisfying $\int_{\mathbb{T}^2} \omega \, d\boldsymbol{x} = 0$
  - Biot-Savart law: $\boldsymbol{u} = \nabla^\perp \Delta^{-1} \omega, \qquad \nabla^\perp = (-\partial_y, \partial_x)$
- An equilibrium solution — the 2D Taylor-Green vortex:

$$\boldsymbol{u}_s = (-\cos(x_1)\sin(x_2), \sin(x_1)\cos(x_2))$$

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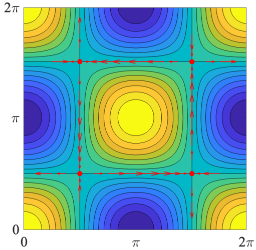

- Linearize the 2D Euler system around a steady solution $(\omega_s, \boldsymbol{u}_s) \Longrightarrow$

$$\begin{aligned} \partial_t w &= \mathcal{L}w, \qquad w\big|_{t=0} = w_0, \qquad \text{where} \\ \mathcal{L}w &:= -(\boldsymbol{u}_s \cdot \nabla)w - (\boldsymbol{u} \cdot \nabla)\omega_s \\ &= \left[-\boldsymbol{u}_s \cdot \nabla - \nabla\omega_s \cdot (\nabla^\perp \Delta^{-2})\right] w, \qquad w \in H_0^m(\mathbb{T}^2) \end{aligned}$$

- Linear stability of the equilibria determined by the spectrum of the operator $\mathcal{L}$

$$\begin{aligned} \sigma(\mathcal{L}) &:= \{\lambda \in \mathbb{C} \ : \ (\mathcal{L} - \lambda) \text{ is not bounded from below in } \mathcal{X}\} \\ &= \sigma_{\text{disc}}(\mathcal{L}) \cup \sigma_{\text{ess}}(\mathcal{L}) \end{aligned}$$

  - A point $z \in \sigma_{\text{disc}}(\mathcal{L})$ if
    (1) z is an isolated point in $\sigma(\mathcal{L})$
    (2) z has finite multiplicity, [illegible]
    (3) The range of $z - \mathcal{L}$ is closed
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- In finite dimensions linear operators can only have point spectrum
- In infinite dimensions the situation is complicated by the presence of the essential spectrum $\sigma_{\mathsf{ess}}(\mathcal{L})$, which may also contain eigenvalues of $\mathcal{L}$ that are not in $\sigma_{\mathsf{disc}}(\mathcal{L})$

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- For the 2D linear Euler operator $\mathcal{L}$ the essential spectrum $\sigma_{\text{ess}}(\mathcal{L})$ is well understood (Friedlander et al. 1997, Shvydkoy & Latushkin 2003)

$$\sigma_{\text{ess}}(\mathcal{L}; H_0^m) = \{z \in \mathbb{C} : |\Re(z)| \le |m|\, \mu_{\max}\},$$

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$$\mu_{\max} := \lim_{t\to\infty} \frac{1}{t} \log \sup_{\boldsymbol{x}\in\mathbb{T}^2} ||\nabla\varphi_t(\boldsymbol{x})||$$

is the maximal Lyapunov exponent corresponding to the Lagrangian flow $\varphi_t : \boldsymbol{\xi} \to \boldsymbol{x}(t;\boldsymbol{\xi})$ generated by the steady state via $\partial_t \boldsymbol{x}(t) = \boldsymbol{u}_s(\boldsymbol{x}(t))$

- If $z \in \sigma_{\text{ess}}(\mathcal{L})$, then there exists a sequence of weak-null unit vectors $\{f_n\} \in H_0^m(\mathbb{T}^2)$, approximate eigenvectors, such that

$$||(\mathcal{L} - z) f_n||_{H^m} \to 0, \qquad n \to \infty$$

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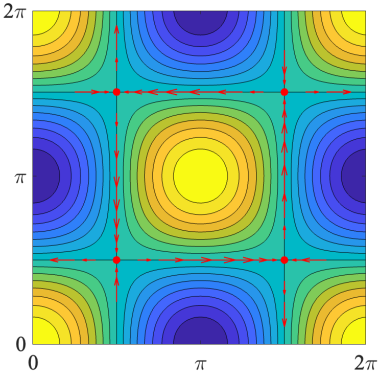

- No general results available about the discrete spectrum $\sigma_{\mathsf{disc}}(\mathcal{L})$
- Unstable eigenvalues found for shear flows (Drazin & Reid 1981; Friedlander et al. 1997), Kolmogorov flows (Friedlander et al. 1997) and the cellular "cat's-eye" flow (Friedlander et al. 2000).
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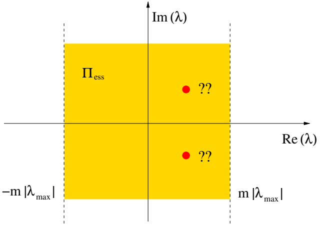

Theorem (Lin 2004)

Suppose there exists an exponentially growing solution $e^{\lambda t} w_0$ *of the linearized system* $\partial_t w = \mathcal{L} w$ *with* $\Re(\lambda) > 0$ *and let* $w_0 \in L^2(\mathbb{T}^2)$. *Then we have the following*

(i) *[regularity of growing modes]* $w_0 \in W^{1,p}(\mathbb{T}^2) \bigcap L^q(\mathbb{T}^2)$ *for all* $1 \le p < p^*$ *and* $1 \le q < \infty$, *where*

$$p^* = \begin{cases} \frac{\mu_{max}}{\mu_{max} - \Re(\lambda)} = \frac{1}{1 - \Re(\lambda)}, & \mu_{max} > \Re(\lambda) \\ \infty, & \mu_{max} \le \Re(\lambda) \end{cases}$$

(ii) *[nonlinear instability] if there exists* $\epsilon > 0$, *such that for any* $\delta > 0$ *there is a solution* $w^\delta(t)$ *of the 2D Euler system corresponding to the initial condition* ω_0^δ, *such that*

$$\|\omega_0^\delta - \omega_s\|_{L^q} + \|\omega_0^\delta - \omega_s\|_{W^{1,p}} \le \delta,$$

and

$$\sup_{0<t<T_\delta} \|\omega^\delta(t) - \omega_s\|_{H^m} \ge \epsilon,$$

where $T_\delta = O(|\ln(\delta)|)$, $q \in [1, \infty)$, $p = [1, p^*)$ *and* $m \ge -1$.

- Numerical solution of the eigenvalue problem $\mathcal{L}\phi(\mathbf{x}) = \lambda\phi(\mathbf{x})$
- Discretize $\mathcal{L}$ using the basis

$$\varphi_{j_1,j_2} = \frac{1}{\sqrt{2}\pi}\cos(j_1 x + j_2 y), \qquad \psi_{j_1,j_2} = -\frac{1}{\sqrt{2}\pi}\sin(j_1 x + j_2 y),$$

$$(j_1, j_2) \in \{(0, j_2), 1 \le j_2 \le N\} \bigcup \{(j_1, j_2), 1 \le j_1 \le N, -N \le j_2 \le N\}$$

to obtain a sparse algebraic eigenvalue problem $\mathbf{L}\,\phi = \lambda\phi$

- The spectrum of $\mathbf{L}$

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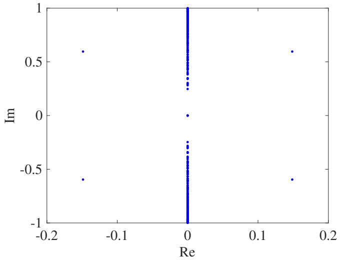

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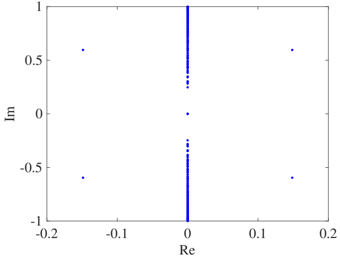

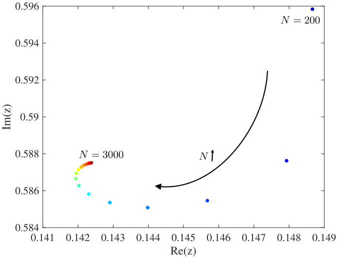

- The unstable eigenvalue: $\lambda_+^{3000} = 0.1424 + 0.5875i$
- The corresponding eigenvectors $\Re\left[\phi_+^N\right]$ are L^2 integrable
- From the theorem of Lin (2004):

$$\phi_+ \in W^{1,p^*}(\mathbb{T}^2), \qquad \text{where} \qquad p^* = \frac{1}{1-0.14} = 1.16$$

- From the Sobolev embedding $W^{1,p^*} \hookrightarrow H^s$ with $1/p^* - 1/2 = 1/2 - s/2$, we have

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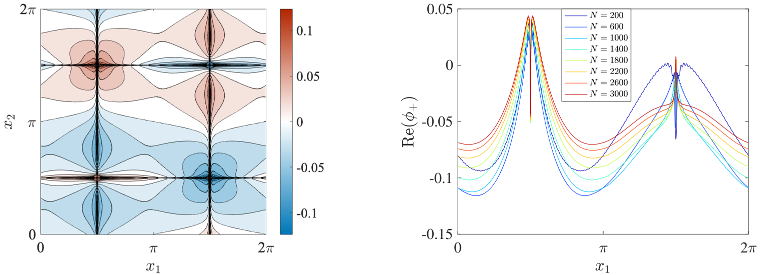

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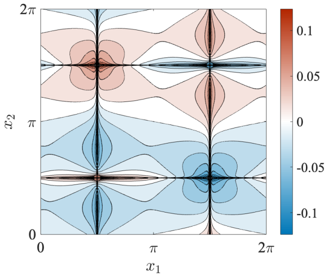

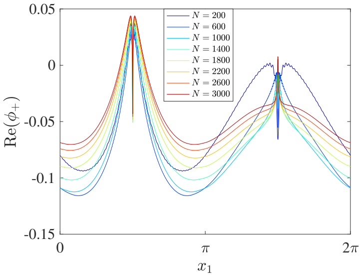

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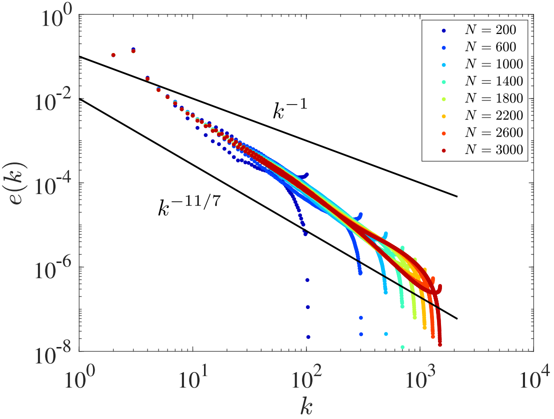
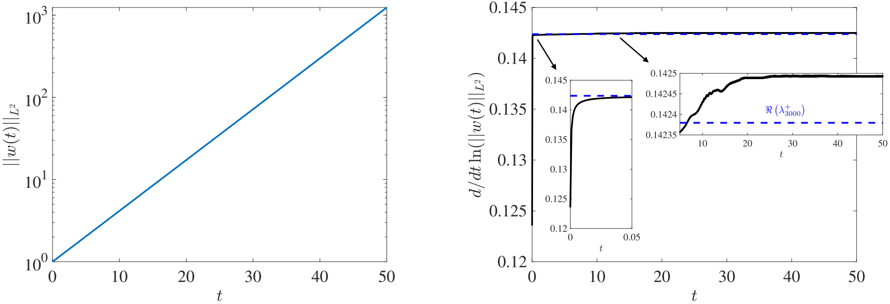
The Fourier spectrum of the eigenfunction ϕ

$$e(k) := \frac{1}{2} \sum_{k \le |\boldsymbol{j}| < k+1} |\widehat{\phi}_{\boldsymbol{j}}|^2, \qquad k \in \mathbb{N}$$

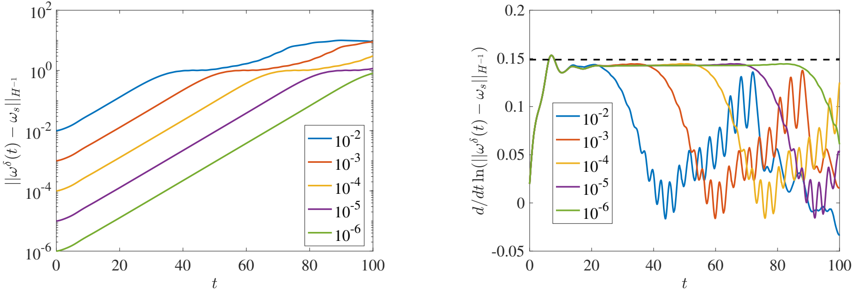


$e(k) = \mathcal{O}\left(k^{-1}\right)$ implies L^2 summability; $e(k) = \mathcal{O}\left(k^{-1.65}\right)$ means $\phi_+ \in H^{0.28}(\mathbb{T}^2)$

Exponential growth of the perturbation



The actual growth rate is within 0.01% equal $\Re(\lambda_+^{3000})$



The time-dependence of (left) the norm $\|\omega^\delta(t) - \omega_s\|_{H^{-1}}$ and (right) the growth rate $(d/dt)\ln(\|\omega^\delta(t) - \omega_s\|_{H^{-1}})$ in the solution of the nonlinear problem with the initial condition given in terms of the eigenfunction ϕ_+^{200} as $\omega_0 = \omega_s + \delta \cdot \Re\left(\phi_+^{200}\right)/||\Re\left(\phi_+^{200}\right)||_{H^{-1}}$ with different indicated magnitudes δ.

- The growth rate of the eigenfunction with $\Re(\lambda_+) \approx 0.1424$ does not saturate the bound on the growth of the semigroup

$$\left\| e^{\mathcal{L}t} w_0 \right\|_{H^m} \le \|w_0\|_{H^m} e^{\mu_{\max} t}, \qquad \mu_{\max} = 1, \quad m = \pm 1$$

- How to find initial data w_0 realizing this growth?

- Given $T > 0$, define the objective functional $J_m : H_0^m(\mathbb{T}^2) \to \mathbb{R}$

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$$\max_{w_0 \in \mathcal{M}} J_m(w_0), \quad \textit{where } m = \pm 1 \textit{ and}$$

$$\mathcal{M} := \{w_0 \in H_0^m(\mathbb{T}^2) : \|w_0\|_{H^m} = 1\}.$$

$$\textit{subject to:} \quad \begin{cases} \partial_t w - \mathcal{L} w = 0, & \textit{in } \mathbb{T}^2 \times (0, T] \\ w = w_0 & \textit{in } \mathbb{T}^2 \textit{ at } t = 0 \\ \textit{Periodic Boundary Conditions} & \end{cases}$$

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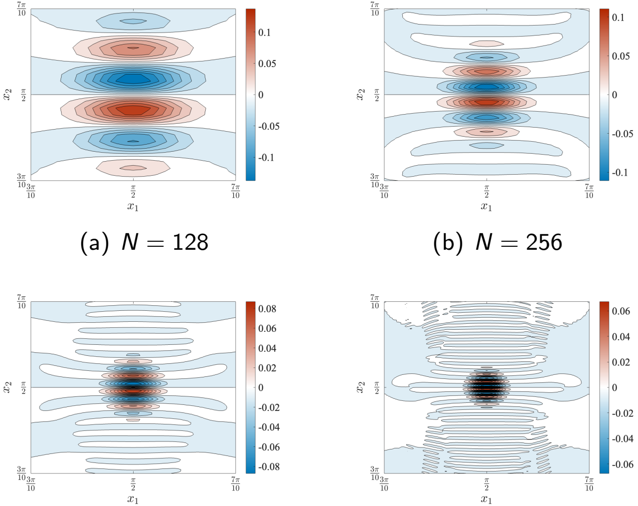


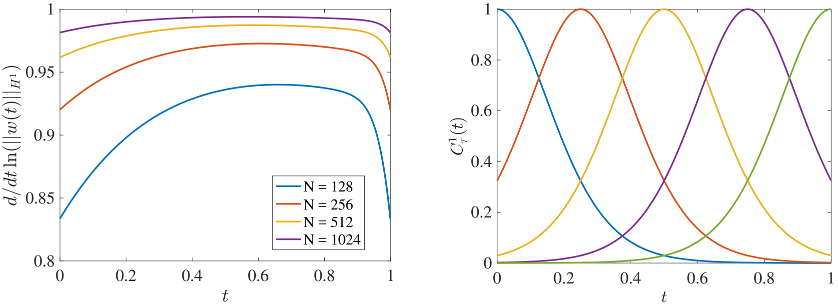
(a) $N = 128$

(b) $N = 256$

(c) $N = 512$

(d) $N = 1024$

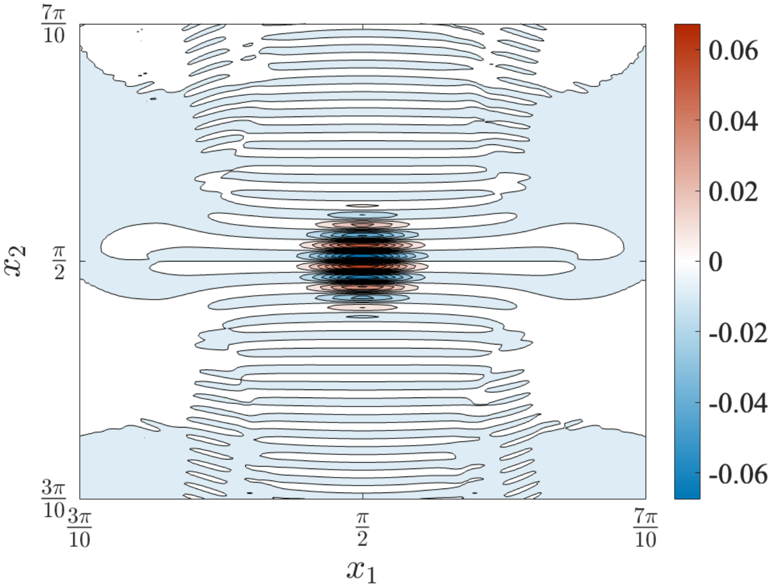
$[m = 1]$ The optimal initial conditions $\widetilde{w}_0^N$ obtained using different resolutions N.



[$m = 1$] (left) The growth rate $(d/dt)\ln(\|w^N(t)\|_{H^1})$ versus t for the optimal initial conditions $\widetilde{w}_0^N$ obtained using increasing spatial resolutions N^2.
(right) The time-dependence of the autocorrelation function

$$\mathcal{C}^1_\tau(t) := \frac{\langle w(\tau), w(t)\rangle_{H^1}}{\|w(\tau)\|_{H^1}\|w(t)\|_{H^1}}, \qquad t, \tau \geq 0$$

corresponding to $N = 1024$ and $\tau = 0, 0.25, 0.5, 0.75, 1$.
Similar results obtained for $m = -1$.


$\frac{7\pi}{10}$
x_2
$\frac{\pi}{2}$
$\frac{3\pi}{10}$
$\frac{3\pi}{10}$
$\frac{\pi}{2}$
$\frac{7\pi}{10}$
x_1
0.06
0.04
0.02
0
-0.02
-0.04
-0.06

Conclusions

- The Taylor-Green vortex is linearly unstable
  - it has a eigenmode given by a distribution with the Sobolev regularity consistent with Lin (2004)
  - it also exhibits a nonmodal growth mechanism involving a continuous family of uncorrelated functions which saturates the spectral bound
    - intrinsically related to the ∞-dimensional nature of the operator $\mathcal{L}$
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